

Muhammad Shehu Abubakar-Sadiq

PhD in Computer Science — Cybersecurity

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PROFESSIONAL SUMMARY

Cybersecurity and digital identity researcher with a PhD in Computer Science and 12+ years of experience across academia, global technology and financial services. Extensive track record in designing secure digital asset management systems, decentralized identity verification, and privacy-preserving data infrastructures. Key contributions include: SPIDVerify, a secure and privacy-preserving decentralised identity verification framework; decentralised real estate transfer verification using Self-Sovereign Identity and smart contracts; Nanostima (€2.5M EU Horizon 2020), the secure health-data infrastructure adopted for Portugal's COVID-19 vaccine tracking; and DECaDE (£6M EPSRC), a UK programme on secure digital economy in supply chain presented to the House of Lords. Hands-on with NIST CSF, ISO/IEC 27001, GDPR, NDPA 2023, risk assessment, DPIAs and secure system architecture. Positioned to support Kano State in building a trustworthy, transparent and auditable Digital Asset Management System (KDAMS).

CORE COMPETENCIES

• Digital Asset Registration & Provenance	• Decentralised Identity & SSI	• Asset Ownership Verification (SPIDVerify)
• Smart Contract & Real Estate Transfer Verification	• Secure Data Infrastructure & Governance	• Risk Assessment (FAIR, ISO 27001, NIST CSF)
• Data Privacy, GDPR & NDPA 2023	• DPIAs, DSARs & Records of Processing	• Incident Response & Breach Reporting
• Privacy by Design & Default	• Fraud Prevention & AML/KYC Controls	• Stakeholder & Cross-Functional Leadership

RELEVANT RESEARCH & PROJECTS

SPIDVerify — Decentralised Identity Verification Framework

2023

Research Lead • University of Porto / INESC TEC

- Designed a secure, privacy-preserving decentralised identity verification framework using Self-Sovereign Identity (SSI).
- Directly applicable to KDAMS asset ownership verification, RBAC authentication and trusted user onboarding across MDAs.
- Published in IEEE SmartNets 2023.

Decentralised Real Estate Transfer Verification

2022

Research Lead • University of Porto / INESC TEC

- Built a decentralised real estate transfer verification system using Self-Sovereign Identity and smart contracts.
- Provides tamper-proof asset ownership records, automated transfer validation and audit trails — directly relevant to KDAMS land and building asset management.
- Published at 19th International Conference on Security and Cryptography (SECRYPT), SCITEPRESS.

Nanostima — Secure Health Data Infrastructure

2017 – 2023

Co-Team Researcher • INESC TEC Porto (€2.5M EU Horizon 2020)

- Contributed to design and security of a large-scale health-data infrastructure platform.
- Developed identity, consent, access-control and audit-logging frameworks for sensitive personal data.
- Platform was adopted as the backbone for Portugal's national COVID-19 vaccine tracking and monitoring system.

DECaDE — Secure Digital Economy in Supply Chain

2024 – 2025

Cyber Security Post-Doctoral Research Fellow • University of Surrey
(£6M EPSRC programme)

- Researched decentralised trust, digital provenance and privacy-preserving reputation for supply chains.
- Designed GDPR-compliant architectures and DPIA-style impact assessments for sensitive data ecosystems.
- Outputs presented to the House of Lords for secure digital economy in supply chain.

PROFESSIONAL EXPERIENCE

Teaching Fellow — Cyber Security

January 2026 – Present

University of Leicester — School of Computing and Mathematical
Sciences

- Research focus on secure digital content provenance, decentralised identity management and GDPR-compliant trust systems.
- Deliver UG/PG teaching in cybersecurity, information security and risk management.

Cyber Security Post-Doctoral Research Fellow

May 2024 – December 2025

University of Surrey — DECaDE Centre (£6M EPSRC programme)

- Designed privacy-preserving and security architectures aligned with GDPR principles and NIST CSF.
- Conducted DPIA-style impact analyses and authored peer-reviewed publications (IEEE EuroS&PW, Springer LNCS).

External Researcher — Information Security & Identity Management

2017 – 2023

INESC TEC Porto (CRACS)

- Contributed to Nanostima (€2.5M EU Horizon 2020) secure health-data infrastructure adopted for Portugal's COVID-19 vaccine tracking.
- Designed identity, consent and access-control flows for sensitive personal data (OAuth, SAML, Shibboleth).
- Aligned architectures to GDPR, NIST and ISO/IEC 27001 security and privacy controls.

Fraud Prevention Consultant — Payments & Customer Data

2020 – 2022

Apple Inc. (via Majorel Portugal)

- Led fraud prevention and detection across all payment channels, protecting millions in assets.
- Triaged VISA CAMS / FICO alerts and produced post-incident reports for Risk Management.

Lecturer II — Information Technology & Cybersecurity

2015 – 2023

Bayero University Kano, Nigeria — Department of Information
Technology

- Designed and delivered modules on cybersecurity, data protection law and IT risk.
- Co-managed faculty computer lab, including access control, asset management and incident handling.

Relationship and Risk Officer

2013 – 2015

Unity Bank PLC, Nigeria

- Conducted KYC, IT risk assessments and cybersecurity control reviews across banking systems.
- Produced risk reports for senior management and delivered internal staff training on data handling and security.

EDUCATION

PhD in Computer Science — Cybersecurity

2017 – 2023

University of Porto, Portugal

Thesis: Establishing Secure and Privacy-Preserving Digital Identity with Self-Sovereign Identity

MSc, Business Information Systems

2011 – 2012

University of East London, United Kingdom**BSc, Computer Science — Digital Systems Security**

2007 – 2011

University of Wollongong, Australia**SELECTED PUBLICATIONS**

Shehu, A.S., Pinto, A., & Correia, M.E. (2023). SPIDVerify: A secure and privacy-preserving decentralised identity verification framework. 2023 International Conference on Smart Applications, Communications and Networking (SmartNets), pp. 1-8. IEEE.

Shehu, A.S., Pinto, A., & Correia, M.E. (2022). A Decentralised Real Estate Transfer Verification Based on Self-Sovereign Identity and Smart Contract. Proceedings of the 19th International Conference on Security and Cryptography (SECRYPT), pp. 646-653. SCITEPRESS.

Shehu, A.S. & Schneider, S. (2025). Decentralised supply chain reputation: A privacy and self-sovereign identity perspective. IEEE European Symposium on Security and Privacy Workshops (EuroS&PW), 474-484.

Fraser, A., Shehu, A.S., Frymann, N., Haynes, P., & Schneider, S. (2025). Privacy-Preserving Photo Sharing: An SSI Use Case. ICISS 2024, Springer LNCS 15416.